

Hesperidin Displays Relevant Role in the Nutrigenomic Effect of Orange Juice on Blood Leukocytes in Human Volunteers: A Randomized Controlled Cross-Over Study

Background We previously showed, in healthy, middle-aged, moderately overweight men, that orange juice decreases diastolic blood pressure and significantly improves postprandial microvascular endothelial reactivity and that hesperidin could be causally linked to the observed beneficial effect of orange juice. The objective was to determine the effect of chronic consumption of orange juice on the gene expression profile of leukocytes in healthy volunteers and to assess to what extent hesperidin is involved in the effect of orange juice.

Methodology/Principal Findings Volunteers were included in a randomized, controlled, crossover study. Throughout three 4-week periods, volunteers consumed daily: 500 ml orange juice, 500 ml control drink plus hesperidin or 500 ml control drink and placebo. Blood samplings were performed on 10 overnight-fasted subjects after the 4-week treatment period. Global gene expression profiles were determined using human whole genome cDNA microarrays. Both orange juice and hesperidin consumption significantly affected leukocyte gene expression. Orange juice consumption induced changes in expression of, 3,422 genes, while hesperidin intake modulated the expression of 1,819 genes. Between the orange juice and hesperidin consumption groups, 1,582 regulated genes were in common. Many of these genes are implicated in chemotaxis, adhesion, infiltration and lipid transport, which is suggestive of lower recruitment and infiltration of circulating cells to vascular wall and lower lipid accumulation.

Conclusions This study shows that regular consumption of orange juice for 4 weeks alters leukocyte gene expression to an anti-inflammatory and anti-atherogenic profile, and hesperidin displays a relevant role in the genomic effect of this beverage. Trial Registration ClinicalTrials.gov NCT 00983086